

A five-variable piecewise cubic function without a polynomial max–min representation

Draft for mathematical review

1 The whole-space function

Let $(a, b, c, d, e) \in \mathbb{R}^5$, and define

$$x = a - 6c + e, \quad y = 4(d - b), \quad z = a + 2c + e, \quad u = 2(e - a), \quad v = 4(d + b), \quad R = 2b + 4d + 2a^2. \quad (1)$$

The first five coordinates are an invertible linear change. Thus this is the actual polynomial graph $R = (y + 3v)/4 + 2((x + 3z - 2u)/8)^2$ over all five original coordinates. Use

$$C = \begin{pmatrix} x & -v + R/2 & -y \\ y & z + u & x \\ z & R/2 & 0 \\ u & -v/2 + R & -v/2 \\ v & 2z + u/2 & u/2 \end{pmatrix}, \quad J = \text{diag}(1, 1, 3, -1, -1). \quad (2)$$

Put $Q = C^T J C$, and let D be the determinant of the first three rows of C . The full polynomial identities are

$$Q = \begin{pmatrix} A_0 & B_0 & 0 \\ B_0 & E_0 - A_0 & F_0 \\ 0 & F_0 & E_0 \end{pmatrix},$$

$$\begin{aligned} A_0 &= 16(ae - 4bd + 3c^2), & B_0 &= 8(2cd - 3be + a^3), \\ E_0 &= 4ae - 12ac - 12ce + 36c^2 + 12d^2 - 40bd + 12b^2, \\ F_0 &= 4ae + 12ac - 20ce - 12c^2 + 12d^2 - 12b^2 + 8bd + 8a^2b. \end{aligned} \quad (3)$$

Define the four original polynomial labels

$$\begin{aligned} q_1 &= -10B_0 + 2E_0 + 2F_0, & q_2 &= 4A_0 + 2B_0 - 2E_0 - 2F_0, \\ q_3 &= -4A_0 + 2B_0 + 6E_0 - 10F_0, & q_4 &= -4A_0 + 2B_0 + 2E_0 + 2F_0. \end{aligned} \quad (4)$$

Let $h = \min(q_1, q_2, q_3, q_4)$, and set

$$f = \begin{cases} h & h > 0 \text{ and } D > 0, \\ 0 & \text{otherwise.} \end{cases} \quad (5)$$

This is a continuous finite-piece polynomial function on all of $\mathbb{R}^5$. For completeness, the fixed simplex reconstruction identity for matrices of the shape (3), with the labels (4), gives

$$h > 0 \implies Q \succeq (h/4)I. \quad (6)$$

One can verify (6) entirely by the following finite matrix identity. Set $s_i = 4e_i - \mathbf{1}$ for $i = 0, 1, 2$, $s_3 = -\mathbf{1}$, $u_i = (e_i + \mathbf{1})/4$ for $i = 0, 1, 2$, and $u_3 = 0$. For $q_{ij} = -s_i^T Q s_j$, the retained edges 01, 03, 12, 13 give q_1, q_2, q_3, q_4 , while

$$\begin{aligned} q_{02} &= 2q_1 + 6q_2 + 5q_4, & q_{23} &= q_1 + 2q_2 + 2q_4, \\ Q &= \sum_{i < j} q_{ij} (u_i - u_j)(u_i - u_j)^T, & \sum_{i < j} (u_i - u_j)(u_i - u_j)^T &= (I + \mathbf{1}\mathbf{1}^T)/4. \end{aligned} \quad (7)$$

If the first three rows of C annihilated a nonzero coefficient vector, its Gram square would be minus the sum of the squares of the remaining two row evaluations. Thus (6) implies $D \neq 0$. The explicit finite closed semialgebraic cover is

$$S_0 = \{D \leq 0\} \cup \bigcup_j \{q_j \leq 0\}, \quad S_i = \{D \geq 0, \ q_j \geq 0 \ \forall j, \ q_i \leq q_j \ \forall j\}, \quad (8)$$

with labels $0, q_1, \dots, q_4$. The pieces cover all of space and agree on overlaps; in particular, on $D = 0$ with nonnegative labels, the minimum is zero. Finite closed pasting proves the asserted continuity.

Theorem. The function (5) is not a finite maximum of finite minima of polynomials in $\mathbb{R}[a, b, c, d, e]$, with arbitrary real coefficients and unrestricted degrees. The invertible change (1) gives the same conclusion in $\mathbb{R}[x, y, z, u, v]$.

The proof below constructs, for every proposed finite collection of leaves, two actual real points contradicting that representation. It does not assume that a trace ideal is a separating ideal.

2 Weighted identities and the base point

Assign weights (5, 6, 7, 8, 9) to (a, b, c, d, e) , respectively, and put

$$S_\lambda(a, b, c, d, e) = (\lambda^5 a, \lambda^6 b, \lambda^7 c, \lambda^8 d, \lambda^9 e). \quad (9)$$

The following three original polynomials are weighted homogeneous:

$$A = ae - 4bd + 3c^2, \quad B = 2cd - 3be + a^3, \quad W = 3d^2 - 4ce + a^2b, \quad (10)$$

with weights 14, 15, 16, respectively. Write

$$g = b^2 - ac, \quad G = -a^2d + 3abc - 2b^3. \quad (11)$$

Their weights are 12 and 18. Directly from (3),

$$\begin{aligned} q_1(S_\lambda v) &= 16\lambda^{14}A(v) - 80\lambda^{15}B(v) + 16\lambda^{16}W(v), \\ q_2(S_\lambda v) &= 48\lambda^{14}A(v) + 16\lambda^{15}B(v) - 16\lambda^{16}W(v), \\ q_4(S_\lambda v) &= -48\lambda^{14}A(v) + 16\lambda^{15}B(v) + 16\lambda^{16}W(v), \end{aligned} \quad (12)$$

and

$$q_3(S_\lambda v) = 192\lambda^{12}g(v) + O(\lambda^{14}), \quad D(S_\lambda v) = 8\lambda^{18}G(v) + O(\lambda^{20}). \quad (13)$$

The remainders in (13) are polynomial in λ and the coordinates of v , and are uniform for v in any fixed bounded set. For the determinant assertion, a full expression is obtained by putting

$$\begin{aligned} h_0 &= -ac - ce + 6c^2 + 2d^2 - 4bd + 2b^2, \quad b_0 = d(-a + 4c + e) + 2b(c - e), \\ D &= 8[(a + 2c + e)b_0 - (b + 2d + a^2)h_0]. \end{aligned} \quad (14)$$

Its weight-18 part is exactly $8G$, and every remaining monomial has weight at least 20.

The monomial trace

$$\tau(t) = (t^5, t^6, t^7, t^8, t^9) \quad (15)$$

satisfies $Q(\tau(t)) = 0$ and $D(\tau(t)) = 0$. We shall use its single real base point

$$p = \tau(1) = (1, 1, 1, 1, 1). \quad (16)$$

Weighted homogeneity, not an assumption of ordinary homogeneity of the selection, is used in (9)–(13).

3 Exact first and second variations

All directions and corrections will have first coordinate zero, so their affine points keep $a = 1$. On this affine hyperplane, A, B, W are quadratic polynomials in the remaining four coordinates. For a direction $v = (0, v_b, v_c, v_d, v_e)$, write their homogeneous quadratic remainders as

$$A_2(v) = -4v_b v_d + 3v_c^2, \quad B_2(v) = 2v_c v_d - 3v_b v_e, \quad W_2(v) = 3v_d^2 - 4v_c v_e. \quad (17)$$

Thus, for example, $A(p + \varepsilon v) = \varepsilon dA_p(v) + \varepsilon^2 A_2(v)$ exactly. Put

$$F = W - B + A, \quad F_2 = W_2 - B_2 + A_2. \quad (18)$$

The full differential dF_p is zero. In particular, $F(p + \varepsilon v) = \varepsilon^2 F_2(v)$ for all the stated directions. Take the explicit rational vectors

$$n = (0, 2, 3, 3, 2), \quad j = (0, 0, 1, 2, 2). \quad (19)$$

They satisfy

$$\begin{aligned} dA_p(n) = dB_p(n) = dW_p(n) = 0, \quad dA_p(j) = dB_p(j) = dW_p(j) = 0, \\ dg_p(n) = 1, \quad dg_p(j) = -1, \quad dG_p(n) = 0, \quad dG_p(j) = 1. \end{aligned} \quad (20)$$

If $F_2(v, w)$ denotes the symmetric polar form with $F_2(v, v) = F_2(v)$, then

$$F_2(n) = 0, \quad F_2(n, j) = 0, \quad F_2(j) = 3. \quad (21)$$

These identities follow immediately from (17): $(A_2, B_2, W_2)(n) = (3, 6, 3)$, $(A_2, B_2, W_2)(j) = (3, 4, 4)$, and their polar values on (n, j) are $(1, 3, 2)$.

Let e_d, e_e denote the corresponding coordinate unit vectors, and use the three-dimensional correction space

$$E = \text{span}(e_d, e_e, j). \quad (22)$$

The first two derivative rows on this basis are

$$\begin{pmatrix} -4 & 1 & 0 \\ 2 & -3 & 0 \end{pmatrix}. \quad (23)$$

Their first two columns have determinant 10; the displayed vectors in (22) are independent because only j has a nonzero c -coordinate.

4 Actual corrected points with matched low-weight polynomials

Fix any $\delta > 0$. Define

$$s(\lambda) = 3 - 3\lambda/8 + \lambda^2, \quad r(\lambda) = \sqrt{s(\lambda)/3}, \quad y_{\pm}(\lambda) = n \pm \delta r(\lambda)j. \quad (24)$$

Here $s(\lambda) = (\lambda - 3/16)^2 + 759/256 > 0$, so the square root is smooth and $r(0) = 1$. For a correction $z \in E$, impose the three real equations

$$\begin{aligned} dA_p(z) + \varepsilon[A_2(y_{\pm}(\lambda) + z) - \lambda^2\delta^2] &= 0, \\ dB_p(z) + \varepsilon[B_2(y_{\pm}(\lambda) + z) - (3\lambda/8)\delta^2] &= 0, \\ F_2(y_{\pm}(\lambda) + z) - \delta^2s(\lambda) &= 0. \end{aligned} \quad (25)$$

At $(\varepsilon, \lambda, z) = (0, 0, 0)$ these equations hold. In the basis (22), the derivative with respect to z has the form

$$\begin{pmatrix} -4 & 1 & 0 \\ 2 & -3 & 0 \\ * & * & \pm 6\delta \end{pmatrix}, \quad (26)$$

whose determinant is $\pm 60\delta \neq 0$. The ordinary real implicit function theorem therefore provides smooth actual solutions $z_{\pm}(\varepsilon, \lambda) \in E$ near $(0, 0)$. For every sufficiently small λ , zero already solves (25) at $\varepsilon = 0$, by (20), (21), (24); local uniqueness gives

$$z_{\pm}(0, \lambda) = 0. \quad (27)$$

Intersect the two parameter neighborhoods and choose a rectangle $|\varepsilon| < \varepsilon_0$, $|\lambda| < \lambda_0$ contained in that intersection. All subsequent choices of ε are made inside this fixed rectangle. Hence the final step $\lambda \rightarrow 0^+$, with ε already fixed, stays in the domain of both actual solution functions.

Define actual real points

$$v_{\pm}(\varepsilon, \lambda) = p + \varepsilon(y_{\pm}(\lambda) + z_{\pm}(\varepsilon, \lambda)). \quad (28)$$

The exact quadratic expansions, (25), and $W = F + B - A$ imply

$$A(v_{\pm}) = \lambda^2\varepsilon^2\delta^2, \quad B(v_{\pm}) = (3\lambda/8)\varepsilon^2\delta^2, \quad W(v_{\pm}) = 3\varepsilon^2\delta^2. \quad (29)$$

There are no omitted Gram equations: only these three polynomials are being prescribed, and the remaining label is checked separately below.

At $\lambda = 0$, smoothness and (20), (27) give

$$g(v_{\pm}(\varepsilon, 0)) = \varepsilon(1 \mp \delta) + O(\varepsilon^2), \quad G(v_{\pm}(\varepsilon, 0)) = \pm\varepsilon\delta + O(\varepsilon^2). \quad (30)$$

Consequently, if $0 < \delta < 1/2$ is fixed and $\varepsilon > 0$ is sufficiently small, both values of g in (30) are positive, and the two values of G have opposite signs. Fix such an ε . For sufficiently small positive λ , put

$$X_{\pm} = S_{\lambda}v_{\pm}(\varepsilon, \lambda). \quad (31)$$

Equations (12), (29) give exactly

$$q_1(X_{\pm}) = 34\lambda^{16}\varepsilon^2\delta^2, \quad q_2(X_{\pm}) = q_4(X_{\pm}) = 6\lambda^{16}\varepsilon^2\delta^2. \quad (32)$$

By (13), (30), and continuity, $q_3(X_{\pm})$ has a positive weight-12 leading coefficient and is eventually larger than the values of q_2, q_4 . The determinant signs are positive at X_+ and negative at X_- . Thus

$$f(X_+) = 6\lambda^{16}\varepsilon^2\delta^2 > 0, \quad f(X_-) = 0. \quad (33)$$

The choices are sequential: δ is chosen first, then a fixed positive ε , and finally a sufficiently small positive λ . All points and equalities in (25)–(33) are over the reals.

5 Every weighted-homogeneous test through weight 16

Let $\mathcal{P}_w$ be the vector space of weighted-homogeneous polynomials of weight w for the weights (9). If $H \in \mathcal{P}_w$, then

$$H(\tau(t)) = t^w H(p). \quad (34)$$

For weights at most 16, all nonzero monomial spaces, apart from the constant, are as follows:

5, 6, 7, 8, 9	a, b, c, d, e respectively	
10, 11	a^2 , ab respectively	
12	ac , b^2	
13	ad , bc	
14	ae , bd , c^2	
15	a^3 , be , cd	
16	a^2b , ce , d^2 .	(35)

Weights 1–4 have no monomials. The list is exhaustive because any fourth factor has weight at least 20, and the only possible triples through 16 are a^3 and a^2b .

When $H(p) = 0$, its coefficients in the appropriate row sum to zero. The kernels and their derivative values on n are given by the following explicit bases:

w	basis for $H(p) = 0$	values of $dH_p(n)$	
12	$b^2 - ac$	1	
13	$bc - ad$	2	
14	$bd - ae$, $c^2 - ae$	3, 4	
15	$be - a^3$, $cd - a^3$	4, 6	
16	$ce - a^2b$, $d^2 - a^2b$	3, 4.	(36)

For every other weight through 16, $H(p) = 0$ forces $H = 0$. It follows from (36) that

$$\{H \in \mathcal{P}_w : H(p) = 0, dH_p(n) = 0\} = \begin{cases} \mathbb{R}A & w = 14, \\ \mathbb{R}B & w = 15, \\ \mathbb{R}W & w = 16, \\ \{0\} & 0 \leq w \leq 16 \text{ otherwise.} \end{cases} \quad (37)$$

For example, $A = 3(c^2 - ae) - 4(bd - ae)$, $B = 2(cd - a^3) - 3(be - a^3)$, and $W = 3(d^2 - a^2b) - 4(ce - a^2b)$. This calculation includes every real weighted-homogeneous polynomial of the stated weights, not just the Gram polynomials.

6 A finite collection of arbitrary polynomial leaves

Let $\mathcal{L}$ be any finite collection of polynomials in $\mathbb{R}[a, b, c, d, e]$, with no restriction on coefficients or degrees. We will choose the parameters in (31) so that

$$P(X_+) \geq f(X_+) \implies P(X_-) > 0 \quad (P \in \mathcal{L}). \quad (38)$$

Decompose each leaf into its finitely many weighted-homogeneous components, $P = \sum_w P_w$. Call a component through weight 16 exceptional if it belongs to the corresponding space on the right of (37), including zero. There are finitely many nonexceptional components through weight 16.

For such a component H , either $H(p) \neq 0$, or $H(p) = 0$ and $dH_p(n) \neq 0$, by (37). In the latter case, choose $\delta > 0$ small enough that

$$dH_p(n + \delta j), \quad dH_p(n - \delta j)$$

are both nonzero and have the sign of $dH_p(n)$. One fixed $0 < \delta < 1/2$ works for all the finitely many components. For $\varepsilon > 0$ sufficiently small, their values at $v_{\pm}(\varepsilon, 0)$ consequently have the same nonzero sign: if $H(p) \neq 0$, their leading term is that constant; otherwise it is $\varepsilon dH_p(n \pm \delta j)$. Choose one such ε also satisfying (30).

The same choice can ensure the following slightly stronger statement when needed. Adding any of the finitely many constants $c\varepsilon^2\delta^2$ arising below to these paired values does not change their signs. Indeed the nonzero leading term is either a fixed constant or is linear in ε ; the added term is quadratic. The constants, components, and δ are all fixed before choosing this final ε .

Now analyze one leaf P . If it has a nonexceptional component through weight 16, let w be the smallest such weight and write $H = P_w$. Every earlier nonzero component is a multiple of A or B , and by (29) its evaluation after (9) has effective order λ^{16} .

If $w < 16$, it follows that

$$\lambda^{-w}P(X_{\pm}) \longrightarrow H(v_{\pm}(\varepsilon, 0)) \quad (\lambda \rightarrow 0^+). \quad (39)$$

The two limits have the same nonzero sign. If they are negative, the hypothesis in (38) is eventually impossible; if they are positive, the conclusion in (38) holds.

If $w = 16$, write the earlier components as αA at weight 14 and βB at weight 15. Formula (29) gives instead

$$\lambda^{-16}P(X_{\pm}) \longrightarrow H(v_{\pm}(\varepsilon, 0)) + (\alpha + 3\beta/8)\varepsilon^2\delta^2. \quad (40)$$

The preceding strengthened choice of ε makes these limits same-signed and nonzero. The same reasoning proves (38).

Finally, suppose all components through weight 16 are exceptional. Write them as $\alpha A, \beta B, \gamma W$ in their respective weights. Then

$$\lambda^{-16}P(X_{\pm}) \longrightarrow c_P\varepsilon^2\delta^2, \quad c_P = \alpha + 3\beta/8 + 3\gamma. \quad (41)$$

If $c_P > 0$, then $P(X_-) > 0$ for sufficiently small positive λ . If $c_P \leq 0$, (33) and (41) give $P(X_+) < f(X_+)$ eventually, including the case $c_P = 0$. Thus (38) also holds in this last case.

All components of weight greater than 16 contribute $o(\lambda^{16})$: the vectors $v_{\pm}(\varepsilon, \lambda)$ stay bounded and converge as $\lambda \rightarrow 0$, and there are only finitely many components of each fixed leaf. No bound on their weights is assumed. Since $\mathcal{L}$ is finite, one sufficiently small positive λ makes (33) and (38) hold simultaneously for every leaf. This completes the unrestricted-degree finite-leaf assertion.

7 Contradiction to a polynomial lattice representation

Suppose that on all of $\mathbb{R}^5$,

$$f = \max_{1 \leq i \leq m} \min_{1 \leq j \leq n_i} P_{ij}$$

for finitely many original real polynomials, with $m, n_i \geq 1$. Apply Section 6 to that finite set of leaves. At X_+ , a row attains the outer maximum $f(X_+) > 0$; every leaf in that row is at least $f(X_+)$. By (38), every one of those finitely many leaves is strictly positive at X_- . Their finite minimum is strictly positive, so the represented value at X_- is strictly positive. This contradicts the actual value $f(X_-) = 0$ in (33).

This proves the theorem for arbitrary real coefficients and arbitrary finite degrees. Any finite expression using minima and maxima can be converted to the displayed form by distributivity, so allowing a different finite lattice syntax does not change the conclusion.

The construction is five-dimensional because its free input is exactly (a, b, c, d, e) , equivalently the five coordinates (x, y, z, u, v) in (1). It does not prove that dimension five is minimal or settle any universal positive statement in dimensions three or four.